CYCLONE DUAL FILTER ELEMENT

The Honda GX390UT2SCM1 engine is equipped with a replaceable, high-density paper air cleaner element. See (Figure 25) for air cleaner components.

1. Follow steps 1 through 3 as outlined in the "Dual Filter Element" section.

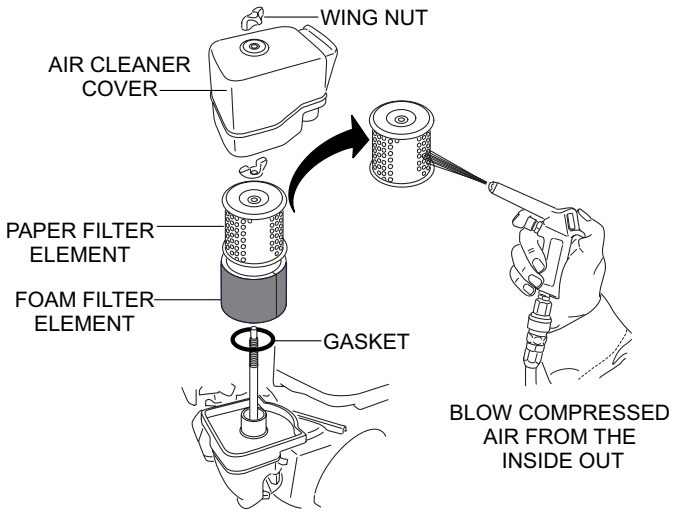


Figure 25. Engine Air Filter (Cyclone)

2. Blow compressed air through the air cleaner cover as shown in Figure 26. Clean inside of air filter cover with warm, soapy water or nonflammable solvent. Rinse and dry thoroughly.

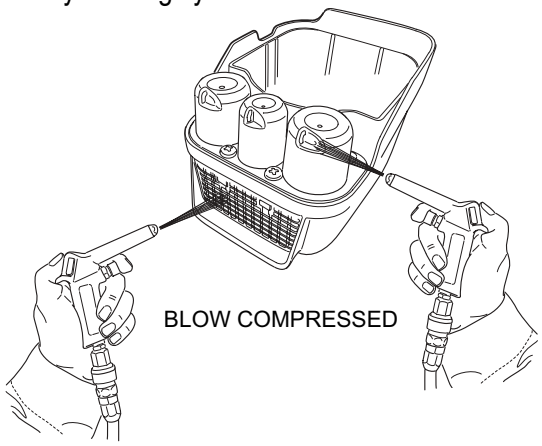


Figure 26. Engine Air Filter Cover

ROLLER INSPECTION AND ADJUSTMENT

1. Refer to Table 8 for oil and grease requirements.
2. Check oil tank level daily.
3. Check the machine for oil leak and proper function of lever, cables and links daily.

MACHINE MAINTENANCE

1. At the end of each day's operation, wash down dust and dirt off the machine. Clean area around drums and scrapers making sure all mud is removed.
2. Drain water tank completely.
3. Cover the machine to prevent dust and store in dry place away from sun exposure.

LONG TERM STORAGE

1. Conduct thorough lubrication and oil change.
2. Disconnect battery terminals and dismount battery from machine. Store battery.
3. If there's a possibility that ambient temperature will drop below freezing point, add antifreeze agent to coolant.
4. Cover the inlet and outlet of air cleaner and muffler securely.
5. Store machine indoors. Do not leave outdoors.
6. Refer to Table 8 for lubrication necessary for the machine.

BATTERY MAINTENANCE

NOTICE

Read and understand the battery safety information in the front of this manual before performing maintenance on the battery.

1. Use a flashlight to check battery electrolyte level. Always check that the engine is stopped.
2. If a battery has not been used for some time, reduce the charge level initially to protect each plate inside the battery.
3. Check the battery terminals periodically to ensure that they are in good condition.

4. Use a wire brush or sand paper to clean the battery terminals.
5. Check battery for cracks or any other damage. If white pattern appears inside the battery or paste has accumulated at the bottom, replace the battery.
6. Measure the specific gravity of electrolyte:
   - completely charged: 1.270 - 1.290
   - needs charging: 1.260 or lower.
7. If the machine will not be in operation for a long period of time, charge the battery sufficiently, tighten all caps correctly, store in cool dry place and check the battery charge level every month to maintain the performance of the battery.

BATTERY CABLE CONNECTION

1. When removing cable, disconnect the ground side (normally negative) first (Figure 27).



Figure 27. Battery Connection

FORWARD AND REVERSE TRAVEL ADJUSTMENT

2. If neutral position for forward and reverse travel has been displaced, conduct the neutral adjustment.
3. If roller travels forward with the ball of ball plunger remaining in V slot of the guide, loosen M8 bolt and slide the slide plate slightly toward the engine. If the roller travels backward, slide the plate towards side plate (Figure 28).

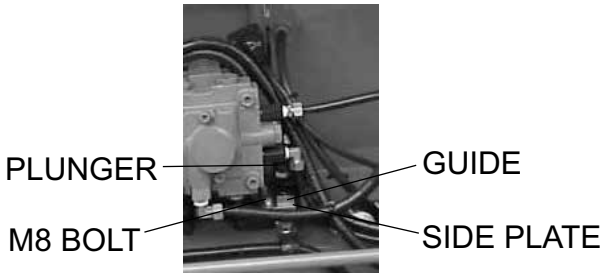


Figure 28. Adjustment of Neutral Position

4. When installing cable, connect the ground side (normally negative) last.
5. With M8 bolt tightened, start engine and check the neutral for forward and reverse. If still displaced, repeat the procedure.
6. If neutral position of forward/reverse lever has been displaced, use the turn buckle located on the oil pump side of forward/reverse cable.

VIBRATION CLUTCH ADJUSTMENT

1. Move vibration lever to the OFF position.
2. Loosen the front and rear nuts at the end of vibration clutch cable (Figure 29).

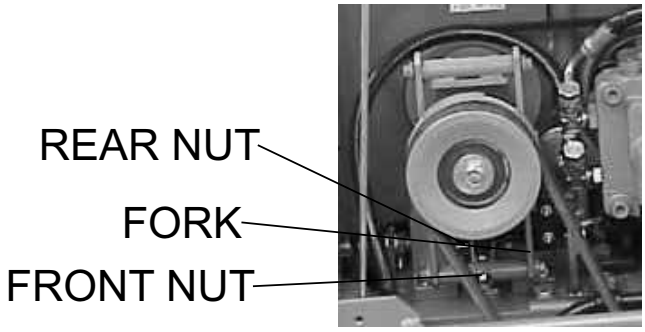
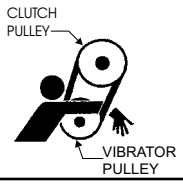


Figure 29. Vibration Clutch Adjustment

3. Turn in the nut on the front side again and at the position that release fork starts to move, turn in the nut by one or two threads. Lock it a this position together with the rear nut.

CAUTION



NEVER attempt to check the V-belt with the engine running. Severe injury can occur if your hand gets caught between the V-belt and the clutch. Always use safety gloves.

CAUTION

Excessive turning of front nut may cause the sliding engagement of clutch to take place or no vibration takes place even when vibration lever is engaged.

On the other hand, insufficient turn may cause sliding engagement of clutch even when the vibration lever is placed in the **OFF** position or for vibration to remain engaged.

HYDRAULIC SYSTEM INSPECTION AND SERVICE

1. Check motor and pump for any damage.
2. Check hoses and pipes for proper tightness and make sure there are no leaks.
3. Check nylon tubes for hydraulic oil intake and drain. Retighten brass nut if loose and if a leak is detected. If leak continues after retightening, replace nylon tube, nut and sleeve.
4. Check oil tank for proper oil level using the hydraulic system oil level gauge (See Figure 6). Make sure hydraulic oil has not whitened or emulsified. Whitish color means aeration in pump. Retighten pipe and correct level of oil. Emulsification means water in the hydraulic oil. Replace the oil.

Table 8. Lubrication Chart			
ITEM	**MAINTENANCE NEEDED**	**FREQUENCY**	**TYPE**
Vibrator Oil	Replace Oil	Every 300 Hours of Operation	SAE10W30 (1.5 liters)
Engine Oil	Replace Oil	Initially - After 10 to 20 Hours of Operation Thereafter - Every 50 to 100 Hours of Operation	SAE#30 or SAE10W30 (Spring - Summer) SAE#20 or SAE10W30 (Fall - Winter) SAE10W30 (Extremely Cold Region)
Hydraulic Oil	Replace Oil	Every 1000 to 1500 Hours of Operation	Viscosity: ISO VG32 equivalent - for cold region ISO VG46 or 56 equivalent - for warm region Shipped from factory with Shell Tellus 46 (25 liters)
Travel Lever, Vibration Lever, Throttle Lever	Add Lubrication Oil to Sliding Parts	Every 50 Hours of Operation	Lubrication Oil
Deadman Device	Lubricate Grease Fitting	Every 50 Hours of Operation	Grease
Handle Bar Release Pin	Lubricate Grease Fitting	Every 50 Hours of Operation	Grease
Bearing Cover	Lubricate Grease Fitting	Every 50 Hours of Operation	Grease
Clutch Box	Lubricate Grease Fitting	Every 50 Hours of Operation	Grease
Travel Lever	Lubricate Sliding Parts	Every 50 Hours of Operation	Grease
Oil Filter	Replace Oil Filter	Initially - After 25 Hours of Operation Thereafter - Every 500 Hours of Operation	Mikasa Genuine 10 micron Filter Paper

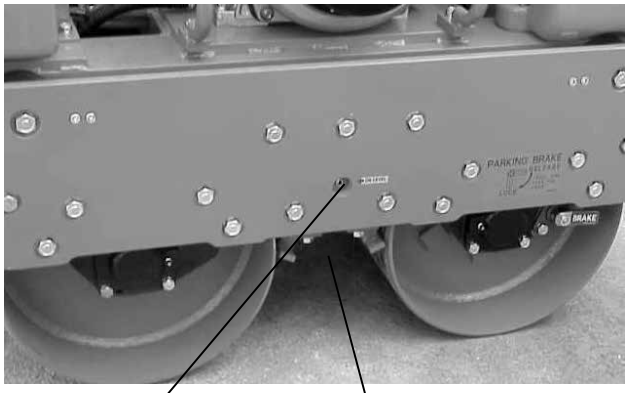
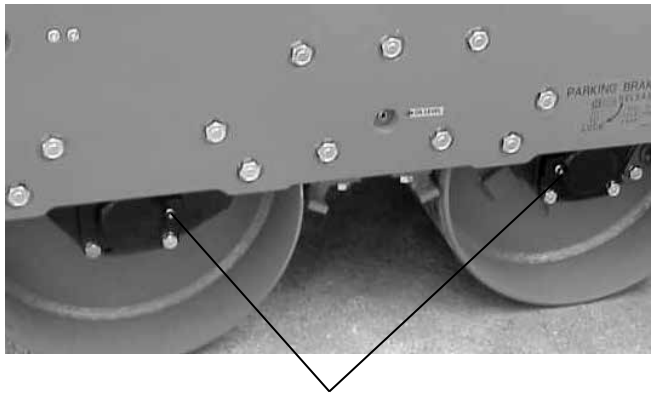


Figure 30. Vibrator Oil Maintenance



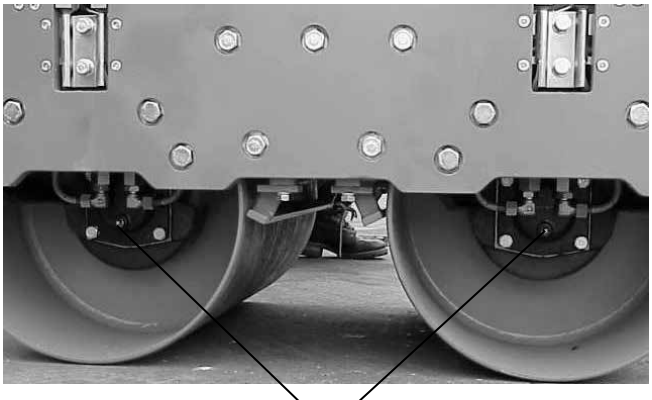


Figure 31. Bearing Cover Maintenance

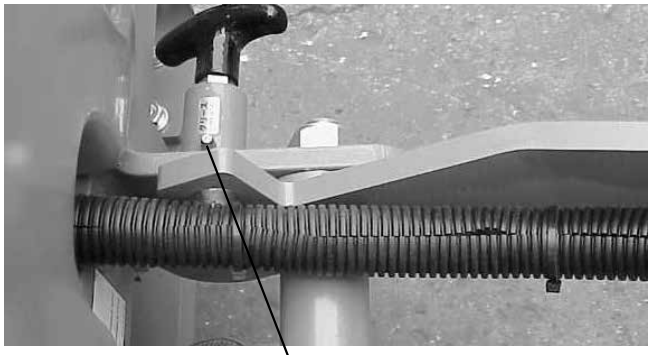


Figure 32. Handle Bar Maintenance

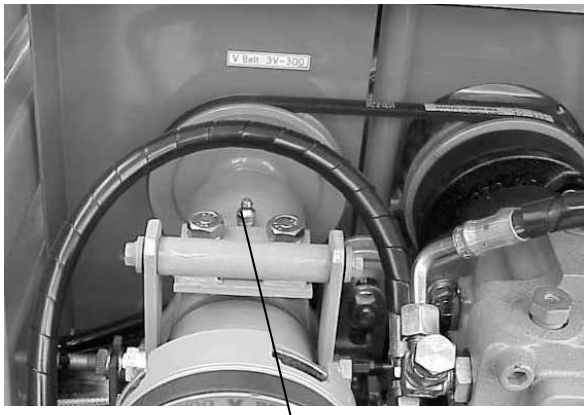


Figure 33. Clutch Box Maintenance



Figure 34. Hydraulic Oil Filter Location

ELECTROMAGNETIC CLUTCH FOR VIBRATION SYSTEM

When the contact area of electromagnetic clutch (Table 9) is worn, it makes the gap slightly bigger between them. If the gap is over 0.028 in (0.7 mm), improper actuation or absorption can occur. Replace electromagnetic clutch at this point.

Table 9. Specifications for Electromagnetic Clutch for Vibration System	
Dead Worn Torque	25 Nm (2.5 kgf-m)
Rated Voltage	12V DC
Coil Resistance at 68 °F (20 °C)	6.6 ohms
Limited Gap	0.028 in (0.7 mm)

In case of over amperage, the fuse (5A) which is built in wire harness, will cut the line automatically. This function does not affect travelling of the roller, at which the engine starts and runs at 2,500 rpm. If there is no vibration, replace the fuse.

Troubleshooting - Roller		
SYMPTOM	**POSSIBLE PROBLEM**	**SOLUTION**
Unit does not travel.	Parking brake still engaged?	Release parking brake lever.
Unit does not travel or travel is not smooth.	Defective centrifugal clutch?	Repair or replace clutch.
	Damaged rubber coupling and flange?	Replace rubber coupling and flange.
	Defective travel cable and link?	Repair or replace travel cable and link.
	Damaged scraper or too much mud in scraper?	Replace or repair scraper.
	Damaged or clogged oil filter?	Replace filter.
	Damaged or leaking hydraulic pipe?	Repair or replace parts.
	Low oil level or contaminated oil?	Replenish or replace oil.
	Damaged or leaking hydraulic pump?	Repair or replace hydraulic pump.
	Damaged or leaking hydraulic motor?	Repair or replace hydraulic motor.
	Damaged drum gear and bearing?	Repair parts.
	Bad drum rotation?	Repair or replace drum.
Unit does not vibrate or has weak vibration.	Defective centrifugal clutch?	Repair or replace clutch.
	Damaged or slipping V-belt?	Replace V-belt or adjust tension.
	Damaged vibration cable and linkage?	Replace or repair vibration cable and linkage.
	Damaged vibration clutch?	Adjust or replace clutch.
	Defective clutch vibrator pulley V-belt?	Replace V-belt.
	Vibrator does not turn smoothly with hand?	Check and repair vibrator. Check if oil level is not excessively high.

Troubleshooting (Engine)		
Symptom	**Possible Problem**	**Solution**
Difficult to start, fuel is available, but no spark at spark plug.	Spark plug bridging?	Check gap, insulation or replace spark plug.
	Carbon deposit on spark plug?	Clean or replace spark plug.
	Short circuit due to deficient spark plug insulation?	Check spark plug insulation, replace if worn.
	Improper spark plug gap?	Set to proper gap.
	Spark plug is red?	Check transistor ignition unit.
	Spark plug is bluish white?	If insufficient compression, repair or replace engine. If injected air leaking, correct leak. If carburetor jets clogged, clean carburetor.
	No spark present at tip of spark plug?	Check if transistor ignition unit is broken, and replace defective unit. Check if voltage cord cracked or broken and replace. Check if spark plug is fouled and replace.
	No oil?	Add oil as required.
	Oil pressure alarm lamp blinks upon starting? (if applicable)	Check automatic shutdown circuit, oil sensor. (if applicable)
Difficult to start, fuel is available, and spark is present at the spark plug.	ON/OFF switch is shorted?	Check switch wiring, replace switch.
	Ignition coil defective?	Replace ignition coil.
	Improper spark gap, points dirty?	Set correct spark gap and clean points.
	Condenser insulation worn or short circuiting?	Replace condenser.
	Spark plug wire broken or short circuiting?	Replace defective spark plug wiring.
Difficult to start, fuel is available, spark is present and compression is normal.	Wrong fuel type?	Flush fuel system, replace with correct type of fuel.
	Water or dust in fuel system?	Flush fuel system.
	Air cleaner dirty?	Clean or replace air cleaner.
	Choke open?	Close choke.
Difficult to start, fuel is available, spark is present and compression is low.	Suction/exhaust valve stuck or protruded?	Reseat valves.
	Piston ring and/or cylinder worn?	Replace piston rings and/or piston.
	Cylinder head and/or spark plug not tightened properly?	Torque cylinder head bolts and spark plug.
	Head gasket and/or spark plug gasket damaged?	Replace head and spark plug gaskets.
No fuel present at carburetor.	No fuel in fuel tank?	Fill with correct type of fuel.
	Fuel cock does not open properly?	Apply lubricant to loosen fuel cock lever, replace if necessary.
	Fuel filter/lines clogged?	Replace fuel filter.
	Fuel tank cap breather hole clogged?	Clean or replace fuel tank cap.
	Air in fuel line?	Bleed fuel line.

Troubleshooting (Engine) - continued		
Symptom	**Possible Problem**	**Solution**
Weak in power, compression is proper and does not misfire.	Air cleaner dirty?	Clean or replace air cleaner.
	Improper level in carburetor?	Check float adjustment, rebuild carburetor.
	Defective spark plug?	Clean or replace spark plug.
	Improper spark plug?	Set to proper gap.
Weak in power, compression is proper but misfires.	Water in fuel system?	Flush fuel system and replace with correct type of fuel.
	Dirty spark plug?	Clean or replace spark plug.
	Ignition coil defective?	Replace ignition coil.
Engine overheats.	Wrong type of fuel?	Replace with correct type of fuel.
	Cooling fins dirty?	Clean cooling fins.
	Intake air restricted?	Clear intake of dirt and debris. Replace air cleaner elements as necessary.
	Oil level too low or too high?	Adjust oil to proper level.
Rotational speed fluctuates.	Governor adjusted incorrectly?	Adjust governor.
	Governor spring defective?	Replace governor spring.
	Fuel flow restricted?	Check entire fuel system for leaks or clogs.
Recoil starter malfunctions. (if applicable)	Recoil mechanism clogged with dust and dirt?	Clean recoil assembly with soap and water.
	Spiral spring loose?	Replace spiral spring.
Starter malfunctions.	Loose, damaged wiring?	Ensure tight, clean connections on battery and starter.
	Battery insufficiently charged?	Recharge or replace battery.
	Starter damaged or internally shorted?	Replace starter.
Burns too much fuel.	Over-accumulation of exhaust products?	Check and clean valves. Check muffler and replace if necessary.
	Wrong spark plug?	Replace spark plug with manufacturer's suggested type.
Exhaust color is continuously white.	Lubricating oil is wrong viscosity?	Replace lubricating oil with correct viscosity.
	Worn rings?	Replace rings.
Exhaust color is continuously black.	Air cleaner clogged?	Clean or replace air cleaner.
	Choke valve set to incorrect position?	Adjust choke valve to correct position.
	Carburetor defective, seal on carburetor broken?	Replace carburetor or seal.
	Poor carburetor adjustment, engine runs too rich?	Adjust carburetor.
Will not start, no power with key ON. (if applicable)	ON/OFF device not activated ON?	Turn on ON/OFF device.
	Battery disconnected or discharged?	Check cable connections. Charge or replace battery.
	Ignition switch/wiring defective?	Replace ignition switch. Check wiring.

OPERATION MANUAL

HERE'S HOW TO GET HELP

PLEASE HAVE THE MODEL AND SERIAL
NUMBER ON-HAND WHEN CALLING

UNITED STATES

Multiquip Inc.

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6141 Katella Avenue Suite 200
Cypress, CA 90630
E-MAIL: mq@multiquip.com
WEBSITE: www.multiquip.com

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Laval, Quebec, Canada H7L 6V3
E-MAIL: infocanada@multiquip.com

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